

Technology Stack

Date	25 June 2026
Team ID	xxxxxx
Project Name	OptiCrop – AI-Based Crop Recommendation System
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Technology Stack Details:

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular, HTML/CSS)	HTML5, CSS3, JavaScript, Bootstrap	HTML, CSS, and JavaScript are used to build a responsive, simple, and user-friendly web interface. They allow farmers to easily enter soil and environmental parameters and view crop recommendations.

2	Backend / Server-Side (e.g., Node.js, Python/Django, Java)	Python (Flask)	Flask is a lightweight Python web framework used to build the backend application. It handles user requests, processes input data, communicates with the trained Machine Learning model, and returns crop recommendations efficiently.
3	Database / Data Storage (e.g., MySQL, MongoDB, PostgreSQL)	Crop Recommendation Dataset (CSV), Joblib Model Files (.pkl)	The Crop Recommendation Dataset is used to train the Machine Learning model. The trained model (model.pkl) and scaler (scaler.pkl) are stored using Joblib and loaded during prediction. No database is required because user data is not permanently stored.
4	Cloud / Hosting / Deployment (e.g., AWS, Azure, Heroku, Docker)	Render	Render provides a reliable cloud platform for deploying Flask applications. It supports automatic deployment from GitHub, continuous updates, and allows users to access the application online from any device

5	Version Control & CI/CD (e.g., GitHub, GitLab, Jenkins)	GitHub	Git is used for version control, while GitHub is used to manage the project repository, track code changes, collaborate, and deploy the application through Render
6	Third-Party APIs/ Other Tools (e.g., Stripe, Twilio, Google Maps)	Scikit-learn, Pandas, NumPy, Joblib, VS Code, Kaggle Crop Recommendation Dataset	Scikit-learn is used to build and evaluate the Machine Learning models. Pandas and NumPy are used for data preprocessing and analysis. Joblib saves and loads the trained model. VS Code is used for development, and the Kaggle Crop Recommendation Dataset provides the training data for crop prediction.